Tutorial 2: Computer Exercises (Chapter 1) — C1 & C2

ECO375 / Wooldridge (Introductory Econometrics)

0. Setup

This handout walks through **Chapter 1 Computer Exercises C1 and C2** using the `wooldridge` package datasets `wage1` and `bwght`.

```
options(repos = c(CRAN="http://cran.rstudio.com"))

if (!require("wooldridge")) install.packages("wooldridge")
```

```
## Loading required package: wooldridge

## Warning: package 'wooldridge' was built under R version 4.3.3
```

```
if (!require("dplyr")) install.packages("dplyr")
```

```
## Loading required package: dplyr

## Warning: package 'dplyr' was built under R version 4.3.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
if (!require("ggplot2")) install.packages("ggplot2")
```

```
## Loading required package: ggplot2

## Warning: package 'ggplot2' was built under R version 4.3.3
```

1. Computer Exercise C1 (data: `wage1`)

We use the data set **WAGE1**.

```r
data("wage1")      # loads a data.frame named wage1
?wage1             # open documentation in RStudio
```

```
## starting httpd help server ... done
```

1.1 What the question is asking (plain English)

You are asked to (i) summarize education and wage in the sample, (ii) adjust the nominal 1976 wage into 2013 dollars using the CPI, and (iii) count men vs. women.

1.2 Solution

(i) Average education; lowest and highest education

```r
educ_mean <- mean(wage1$educ)
educ_min  <- min(wage1$educ)
educ_max  <- max(wage1$educ)

educ_mean; educ_min; educ_max
```

```
## [1] 12.56274
```

```
## [1] 0
```

```
## [1] 18
```

```r
summary(wage1$educ)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   12.00   12.00   12.56   14.00   18.00
```

A quick plot helps you see the distribution:

```r
ggplot(wage1, aes(x = educ)) +
  geom_histogram(binwidth = 1) +
  labs(title = "Education (years) in WAGE1", x = "Years of education", y = "Count")
```

(ii) Average hourly wage; does it seem high or low?

```
wage_mean_1976 <- mean(wage1$wage)
wage_mean_1976
```

```
## [1] 5.896103
```

```
summary(wage1$wage)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.530   3.330   4.650   5.896   6.880  24.980
```

Interpretation note: this is **in 1976 dollars**, so it will look "low" by modern standards.

(iii) Find CPI for 1976 and 2013

The exercise asks you to obtain CPI values from an outside source (internet/printed). For reproducibility in this tutorial, we plug in commonly-used CPI-U values (base period 1982–1984 = 100):

```
cpi_1976 <- 56.9
cpi_2013 <- 232.957
cpi_1976; cpi_2013
```

```
## [1] 56.9
```

```
## [1] 232.957
```

(iv) Convert average wage into 2013 dollars; does it seem reasonable?

To convert **1976 dollars → 2013 dollars**:

$$\text{wage}_{2013} = \text{wage}_{1976} \times \frac{\text{CPI}_{2013}}{\text{CPI}_{1976}}.$$

```r
wage1 <- wage1 %>%
  mutate(wage_2013 = wage * (cpi_2013 / cpi_1976))

wage_mean_2013 <- mean(wage1$wage_2013)
wage_mean_2013
```

```
## [1] 24.13951
```

```r
summary(wage1$wage_2013)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    2.17   13.63   19.04   24.14   28.17  102.27
```

(v) How many women? How many men?

In `wage1`, `female` is a dummy variable: 1 = female, 0 = male.

```r
n_female <- sum(wage1$female == 1, na.rm = TRUE)
n_male   <- sum(wage1$female == 0, na.rm = TRUE)
n_total  <- nrow(wage1)

c(n_female = n_female, n_male = n_male, n_total = n_total)
```

```
## n_female   n_male  n_total
##      252      274      526
```

```r
table(wage1$female)
```

```
##
##   0   1
## 274 252
```

2. Computer Exercise C2 (data: `bwght`)

We use the data set **BWGHT** (birth weight / smoking / family background).

```r
data("bwght")      # loads a data.frame named bwght
?bwght
```

2.1 What the question is asking (plain English)

You are asked to compute counts and averages related to smoking during pregnancy, interpret why a mean can be misleading when many values are zero, handle missing values, and convert family income units.

2.2 Solution

(i) Sample size; how many smoked during pregnancy?

```r
n <- nrow(bwght)
n_smokers <- sum(bwght$cigs > 0, na.rm = TRUE)
n_nonsmokers <- sum(bwght$cigs == 0, na.rm = TRUE)

c(n_total = n, n_smokers = n_smokers, n_nonsmokers = n_nonsmokers)
```

```
##      n_total    n_smokers n_nonsmokers
##         1388          212         1176
```

(ii) Average cigarettes per day; is the mean a good "typical" measure?

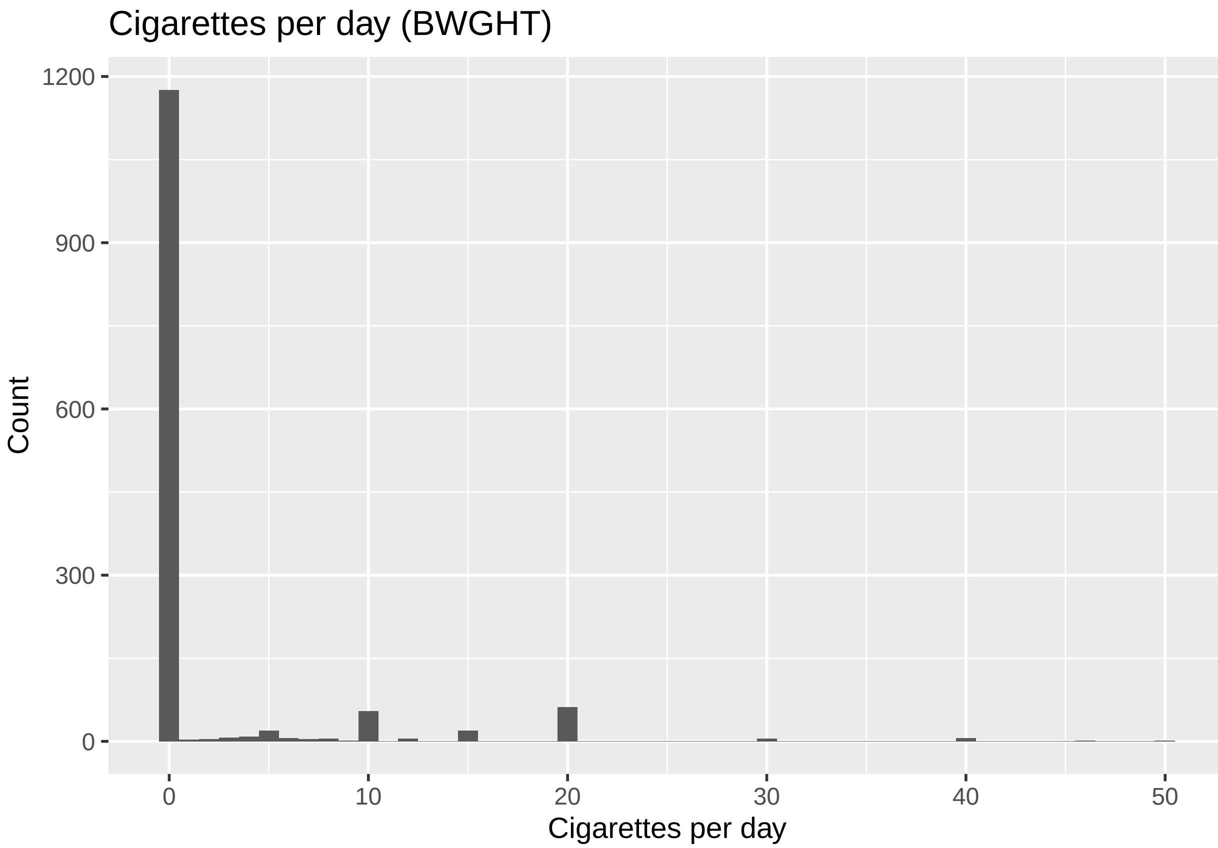
```r
mean_cigs <- mean(bwght$cigs, na.rm = TRUE)
median_cigs <- median(bwght$cigs, na.rm = TRUE)

c(mean_cigs = mean_cigs, median_cigs = median_cigs)
```

```
##   mean_cigs median_cigs
##    2.087176    0.000000
```

A quick diagnostic plot:

```r
ggplot(bwght, aes(x = cigs)) +
  geom_histogram(binwidth = 1) +
  labs(title = "Cigarettes per day (BWGHT)", x = "Cigarettes per day", y = "Count")
```



Interpretation note: when a large share of observations are **zero**, the mean gets "pulled down" and may not represent a "typical" person. In that setting, the **median** (often 0 here) is a better summary of the typical value.

(iii) Average cigarettes per day among smokers; compare and explain

```
mean_cigs_smokers <- mean(bwght$cigs[bwght$cigs > 0], na.rm = TRUE)
mean_cigs_smokers
```

```
## [1] 13.66509
```

This will be much larger than the overall mean because the overall mean averages in a big mass of zeros.

(iv) Average father's education; why only 1,192 observations?

```
fatheduc_mean <- mean(bwght$fatheduc, na.rm = TRUE)
n_used_fatheduc <- sum(!is.na(bwght$fatheduc))
n_missing_fatheduc <- sum(is.na(bwght$fatheduc))

c(mean_fatheduc = fatheduc_mean,
  n_used = n_used_fatheduc,
  n_missing = n_missing_fatheduc,
  n_total = nrow(bwght))
```

```
## mean_fatheduc        n_used     n_missing       n_total
##      13.18624    1192.00000     196.00000    1388.00000
```

Explanation: only observations with **non-missing** values of `fatheduc` can be used in computing the mean.

(v) Average family income and its standard deviation in dollars

In `bwght`, `faminc` is measured in **thousands of dollars**. Convert to dollars by multiplying by 1,000.

```
faminc_dollars <- bwght$faminc * 1000

mean_faminc <- mean(faminc_dollars, na.rm = TRUE)
sd_faminc   <- sd(faminc_dollars, na.rm = TRUE)

c(mean_faminc = mean_faminc, sd_faminc = sd_faminc)
```

```
## mean_faminc   sd_faminc
##    29026.66    18739.28
```

3. Quick checklist (what to turn in)

- C1: mean/min/max education; mean wage (1976); CPI values; mean wage (2013 dollars); counts of women/men.
- C2: sample size and smoker counts; mean and median `cigs`; mean `cigs` among smokers; mean `fatheduc` and explain missingness; mean and SD of family income in dollars.